

A Theory of the Boundaries of Banks with Implications for Financial Integration and Regulation

- Online Appendix -

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1 Market for Retail Deposits

In the first part of this Appendix we consider the case with non-integrated banks but allow for a reallocation of funds through retail competition, next to interbank lending. We show that still our main comparative insights for the size of interbank lending survive. Note that clearly the analysis with an integrated bank would not be affected. As noted in the main text, the fact that integration then eliminates competition on the retail deposit market would generate an advantage for integration, which would blur our analysis.

Model of the Retail Deposit Market. We now allow for retail competition in stage $t = 1$ and show that key results are robust to a potential reallocation of funds via the retail deposit market. It is convenient, however, to first consider the case where only the retail market is active as there is no subsequent interbank lending market: $W = 0$ and thus $F_n = R_n$. This analysis allows us to isolate some key features of how the retail deposit market works in our model. We then solve the model where both retail competition and interbank lending interact.

If a household residing in the local market of bank n deposits not with the local bank but instead with bank n' , it incurs a switching cost $s \geq 0$. For each depositor the respective value of s represents an independent random draw from the cumulative distribution function $G(s)$. The assumption of switching costs that are non-negligible relative to their savings reflects the small granularity of retail deposit financing. Given $M_A \geq M_B$, it is intuitive that in equilibrium switching will occur at most out of market A . For depositors in market A we can then determine a critical switching cost level, s^* , so that only depositors with draws $s \leq s^*$ take up the offer of the rival bank B . If interior, with deposit insurance this yields $s^* = r_B - r_A$ and without deposit insurance $s^* = p(r_B - r_A)$. The respective attracted deposit volumes are then given by $R_A = M_A[1 - G(s^*)]$ and $R_B = M_B + M_A G(s^*)$.

If both banks choose an interest rate above the participation constraint of depositors, $r_n = 0$ or $r_n = 1/p - 1$ for the cases with and without deposit insurance, then these are pinned down by the respective first-order conditions

$$\frac{dR_n}{dr_n} [L'(R_n) - (1 + r_n)] - R_n = 0. \quad (\text{A1})$$

Here, we have $\frac{dR_n}{dr_n} = pg(s^*)M_A$ without deposit insurance and $\frac{dR_n}{dr_n} = g(s^*)M_A$ with deposit insurance. We restrict attention to cases where $g(0)$ is sufficiently large to ensure that the equilibrium deposit rates are characterized by the respective conditions (A1).¹ In order to facilitate the exposition, we stipulate a uniform distribution $s \in [0, \bar{s}]$ with $g(s) = 1/\bar{s}$ and thus suppose that $\bar{s}$ is not too large.

As is immediate, competition will not fully bridge the differences in the deposit base. Formally, this is most immediately seen from the first-order conditions (A1): Given that $\frac{dR_A}{dr_A} = \frac{dR_B}{dr_B}$ and given that at $r_A = r_B$ it holds that $R_A = M_A > R_B = M_B$, these can indeed only jointly hold for bank A and bank B when $r_B > r_A$ though still $R_A > R_B$. The reason for this is the low granularity of retail deposits combined with inertia, as modeled by switching costs. There is, however, an interesting twist to this observation when we now briefly compare the outcomes with and without deposit insurance. We find that the resulting allocation remains more asymmetric when there is no deposit insurance than when there is deposit insurance. Essentially, deposit insurance leads to a decrease in the cost of attracting deposits. This intensifies competition and ensures that the outcome more closely reflects the different marginal profitability in loan making at the two banks, depending on the attracted and invested funds.

Proposition A1. *Suppose (only for now) that interbank loans are not possible, but that there is competition for retail deposits across local markets, albeit hampered by frictions due to switching costs. Then, a difference in the size of the deposit base, $M_A > M_B$, still leads to different volumes of attracted deposits, $R_A > R_B$, and there is less reallocation of funds across markets (larger $R_A - R_B$) under deposit insurance than without deposit insurance.*

¹A corner solution may arise when, starting from the monopoly interest rates, an increase will induce instead too little switching.

Proof. It is now convenient to set up the banks' problem slightly differently for the proof. We suppose that banks compete for depositors by promising some value v_n , so that $v_n = p_n(1+r_n)$ without deposit insurance and $v_n = (1+r_n)$ with insurance. Note that we can then always express the cutoff as $s^* = v_B - v_A$, provided that we still restrict wlog the analysis to the case where $M_A \geq M_B$. Without deposit insurance the first-order condition w.r.t. v_n is obtained from maximizing

$$\pi_n = pL(R_n) - v_n R_n$$

and thus equal to

$$pL'(R_n) \frac{dR_n}{dv_n} - R_n - v_n \frac{dR_n}{dv_n} = 0.$$

With deposit insurance we have, instead,

$$\pi_n = pL(R_n) - pv_n R_n$$

and thus

$$pL'(R_n) \frac{dR_n}{dv_n} - pR_n - pv_n \frac{dR_n}{dv_n} = 0.$$

Note next that

$$\frac{dR_n}{dv_n} = M_A g(s^*) = M_A \frac{1}{s},$$

making use also of the uniform distribution of switching costs. If we now subtract the first-order condition for A from that for B , we obtain without deposit insurance

$$pM_A \frac{1}{s} [L'(R_B) - L'(R_A)] = (R_B - R_A) + M_A \frac{1}{s} (v_B - v_A), \quad (\text{A2})$$

where, as a function of $v_B - v_A$, the left-hand side is strictly decreasing and the right-hand side strictly increasing, noting that $R_B - R_A$ is strictly increasing in $v_B - v_A$ and that $L'' < 0$.

The only change when there is deposit insurance is that the right-hand side of (A2) is multiplied by p , which yields

$$pM_A \frac{1}{s} [L'(R_B) - L'(R_A)] = \left[(R_B - R_A) + M_A \frac{1}{s} (v_B - v_A) \right] p. \quad (\text{A3})$$

Starting from the equilibrium without deposit insurance, where $R_B < R_A$ so that both sides are strictly positive, and multiplying the right-hand side by $p < 1$, to restore equality so as

to obtain (A3), we must increase $v_B - v_A$ and thus increase $R_B - R_A$, as asserted in the Proposition.² **Q.E.D.**

Reallocation of Funds through Both the Deposit Market and Interbank Lending.

Suppose now again that interbank lending is feasible, though in contrast to the baseline analysis, funds can now also be reallocated through the retail deposit markets. We presently stipulate that there is full deposit insurance (covering, for concreteness, also the promised interest rate r_n). Also, we focus again on the case where $\bar{s}$ is not too large, so that there is indeed competition in equilibrium.

How does the operation of a retail deposit market and interbank lending interact? Suppose first that in equilibrium $W = W_1^* > 0$. Hence, if this case prevails, a reallocation of funds is indeed obtained both through the retail and through the interbank channel. The first thing to note is that when $W^* > 0$, the final allocation of funds, as given by F_A and F_B , does not depend on the outcome of retail competition. Still, even though this does not affect the final allocation in this case, banks have an incentive to acquire a larger fraction of the total retail deposit market. We obtain from the respective first-order conditions for $t = 1$ the requirements

$$\begin{aligned}
\frac{d\pi_A}{dr_A} \frac{1}{p} &= [(L'(R_A) - (1 + r_A)) \frac{dR_A}{dr_A} - R_A] \\
&\quad + \frac{1}{p} [(p^2 + \rho p(1 - p)) L'(R_B) - p L'(R_A)] \frac{dR_A}{dr_A} \\
&= 0, \\
\frac{d\pi_B}{dr_B} \frac{1}{p} &= [L'(R_B) - (1 + r_B)] \frac{dR_B}{dr_B} - R_B \\
&= 0.
\end{aligned} \tag{A4}$$

We stipulate that the respective problems are strictly quasiconcave so that the best responses are uniquely determined. In both expressions in (A4) the respective (first) term in rectangular brackets describes the first-order condition when there is no subsequent interbank lending. The second term in bank A 's first-order condition captures the profits that bank A

²This result can also be established by using (A3) and implicitly differentiating $(v_B - v_A)$, obtaining thus that $d(v_B - v_A)/dp < 0$.

will extract from an interbank loan. Notably, using the first-order condition for $W = W_1^*$ this term is indeed strictly positive. This makes bank A relatively more aggressive in the retail deposit market, compared to the benchmark situation where an interbank loan was exogenously ruled out.³ As a consequence, the anticipation of interbank lending reduces the reallocation of funds through retail deposit competition. Interestingly, this will be even more pronounced when the efficient allocation applies. Then, by rearranging the respective first-order condition for bank A we obtain

$$\begin{aligned} \frac{d\pi_A}{dr_A} \frac{1}{[p^2 + \rho p(1-p)]} &= [(L'(R_A) - (1 + r_A)) \frac{dR_A}{dr_A} - R_A] \\ &\quad + [L'(R_B) - L'(R_A)] \frac{dR_A}{dr_A} \\ &= 0. \end{aligned} \tag{A5}$$

The first-order condition for bank B remains the same. The difference between (A4) and (A5) is that the repayment of an interbank loan from bank B is now no longer weighted by $[p + \rho(1-p)]$ in the first-order condition. This reflects the fact that bank A 's shareholders benefit more from a marginal increase in the interbank loan when this leads to contagion as then there is no co-insurance externality on its depositors. Consequently, compared to Case 1, the incentives for bank A to attract deposits - relative to the incentives of bank B - further increase in Case 2. As in Case 2 the overall reallocation of funds increases compared to Case 1, as then $F_A = F_B$ prevails, and as we have noted that there will be less reallocation of funds through retail deposit competition, the size of interbank lending is then higher.

Proposition A2. *The insights on interbank lending from the main analysis still apply when funds can be reallocated also through competition in the retail deposit market, as long as switching costs are not too low. Precisely, the size of interbank lending W^* is still increasing in both the difference in the deposit base (z) and the correlation of lending markets (ρ). Furthermore, W^* again jumps upwards when (at the respective levels of z or ρ) the interbank loan becomes sufficiently large to be contagious.*

³We can show that when the terms of the interbank loan are determined by symmetric Nash bargaining (cf. Part 2 of this Appendix), then this observation still holds for bank A , while then, in addition, bank B 's incentives to attract funds through the retail deposit market are muted. Taken together, the subsequently reported results then still hold.

Proof. Recall from the preceding analysis that the reallocation via retail competition becomes (weakly) smaller if the possibility of interbank lending is introduced. Observe now that if switching costs (precisely, the upper support $\bar{s}$) are sufficiently high satisfying

$$\bar{s} \geq \frac{M+z}{M-z} [L'(M-z) - 1],$$

neither bank A nor bank B offers a deposit rate that exceeds the monopoly rate of zero even when interbank lending is possible. Then reallocation will take place only via interbank lending.

Next, recall the benchmark case where interbank lending is not possible and note that for intermediate values of $\bar{s}$, namely

$$L'(M+z) - 1 \leq \bar{s} < \frac{M+z}{M-z} [L'(M-z) - 1],$$

it will become optimal for bank B to attract some of bank A 's depositors by offering $r_B > 0$, while still $r_A = 0$. Then Case 1 may arise whenever the allocation that is achieved in the benchmark setting without interbank lending is such that

$$(p^2 + p\rho(1-p)) L'(R_{B0}) > pL'(R_{A0}). \quad (\text{A6})$$

We denote the resulting profits for A when it abstains from interbank lending by π_{A0} . Using that for $r_A = 0$ we have that $r_{B1} = r_{B0} = r_B$, we have

$$\begin{aligned} \frac{d}{dz} (\pi_{A1} - \pi_{A0}) &= p[(p + \rho(1-p)) L'(R_B) - L'(R_A)] \left[(1 - G(r_B)) - R' \frac{dr_B}{dz} \right] \\ &> 0, \end{aligned}$$

which follows from implicit differentiation of the first order condition for B :

$$\begin{aligned}
\left[(1 - G(r_B)) - R' \frac{dr_B}{dz} \right] &= \frac{(1 - G(r_B)) - [L'(R_B) - (1 + r_B)] / \bar{s}}{2 - L''(R_B) R'} \\
&= \frac{(1 - G(r_B)) - R_B / (\bar{s} R')}{2 - L''(R_B) R'} \\
&= \frac{2(R_A - M)}{(M + z) [2 - L''(R_B) R']} \\
&> 0.
\end{aligned} \tag{A7}$$

Likewise, differentiating $\pi_{A1} - \pi_{A0}$ with respect to ρ and using the first order condition for W in Case 1 yields

$$\begin{aligned}
\frac{d\pi_{A1}}{d\rho} &= p(1 - p)w - [(p^2 + \rho p(1 - p)) L'(R_B) R' - pR] \frac{\partial r_B}{\partial \rho} \\
&= p(1 - p)w \\
&> 0,
\end{aligned}$$

which follows from the observation that $\frac{\partial r_B}{\partial \rho} = 0$ when $r_A = 0$. To pin down the transition to Case 2, consider again

$$\frac{d}{d\rho} (\pi_{A2} - \pi_{A0}) = p(1 - p) [2L(M) - L(R_B) - R_A].$$

Clearly, for $z = 0$ we have $r_B = 0$ such that $R_A = R_B = M$, implying that this expression is positive due to assumption (1). Differentiating with respect to z yields

$$p(1 - p) (L'(R_B) - 1) (1 - G(r_B)),$$

which is strictly positive due to assumption (1). Differentiating $\pi_{A2} - \pi_{A1}$ and noting again that $r_{B2} = r_{B1} = r_B$ for $r_A = 0$ and using the first order condition for W_1^* yields

$$\frac{d}{d\rho} (\pi_{A2} - \pi_{A1}) = p(1 - p) [2L(M) - L(F_{B1}) - R_A].$$

As in the setting without retail competition, it suffices to show that this is positive for the

highest F_{B1} that is still compatible with Case 1. The sign of the derivative is then determined by

$$2L(M) - L(F_{A1}) - L(F_{B1}) > 0.$$

With respect to variations in z , we have

$$\frac{d}{dz}(\pi_{A2} - \pi_{A0}) = \left[(p^2 + \rho p(1-p)) L'(R_B) - pL'(R_A) \right] \left[(1 - G(r_B)) - R' \frac{\partial r_B}{\partial z} \right]$$

which is strictly greater than zero for $[p^2 + \rho p(1-p)] L'(R_B) > pL'(R_A)$ due to (A7). Now consider

$$\frac{d}{dz}(\pi_{A2} - \pi_{A1}) = p(1-\rho)(1-p) \left[(1 - G(r_B)) - R' \frac{\partial r_B}{\partial z} \right],$$

which is strictly positive due to (A7). Finally, for

$$\bar{s} < L'(M+z) - 1,$$

both r_B and r_A are positive and interbank lending will never arise, even if $pL'(R_{A0}) - (p^2 + \rho_1 p(1-p)) L'(R_{B0}) < 0$. To see this, consider the point where

$$(p^2 + \rho p(1-p)) L'(R_{B0}) = pL'(R_{A0}),$$

and thus $\pi_{A1} - \pi_{A0} = 0$. Since $\frac{d\pi_{A0}}{d\rho} = 0$, it suffices to consider

$$\begin{aligned} \frac{d\pi_{A1}}{d\rho} &= p(1-p) [L(F_{B1}) - L(R_{B1})] - pR_{A1} \frac{\partial r_{B1}}{\partial \rho} \\ &\quad + [(p^2 + \rho p(1-p)) L'(F_{B1}) - pL'(F_{A1})] \frac{\partial F_{B1}}{\partial \rho} \\ &\quad - [(p^2 + \rho p(1-p)) L'(R_{B1}) - p(1+r_{A1}) - R_{A1}] R' \frac{\partial s^*}{\partial \rho} \\ &= p(1-p)w - pR_{A1} \frac{\partial r_{B1}}{\partial \rho}, \end{aligned}$$

which is strictly negative at the point where $w = 0$, since $\frac{\partial r_{B1}}{\partial \rho} > 0$. The latter follows from

implicit differentiation of the first-order condition for B , which yields

$$\frac{\partial r_{B1}}{\partial \rho} = [L''(R_{B1}) R' - 1] \frac{\partial s^*}{\partial \rho}.$$

The derivative $\frac{\partial s^*}{\partial \rho}$ is strictly negative, which follows from adding up the two first order conditions (A4), giving

$$\begin{aligned} y &= R' s^* - (R_A - R_B) + R'[L'(R_A) - L'(R_B)] \\ &\quad + R' \frac{1}{p} [(p^2 + \rho p(1-p)) L'(R_B) - p L'(R_A)] \\ &= 0, \end{aligned} \tag{A8}$$

and implicit differentiation, which yields

$$\begin{aligned} \frac{\partial s^*}{\partial \rho} &= \frac{L'(R_{B1})}{R'(1-\rho)p(1-p)L''(R_{B1}) - 3p} \\ &< 0. \end{aligned}$$

Q.E.D.

Proposition A2 thus confirms that our previous comparative analysis for the size of inter-bank lending is robust also to the operation of a market for retail deposits. This is intuitive given that the market for retail deposit does not provide an adequate substitute due to a combination of switching costs and the low granularity of individual deposits.⁴ Finally, also with regard to its regulatory implications, it is worthwhile to stress the following result:

Proposition A3. *When interbank lending is feasible, then even though there is retail deposit competition it will always (weakly) increase the extent to which funds are reallocated between the two local markets and thereby improve efficiency. That is, when $z > 0$ and thus $M_A > M_B$, interbank lending will reduce or even fully close the gap $F_A - F_B > 0$ that persists also*

⁴The preceding discussion as well as the proof of Proposition A2 entail in addition implications that relate directly to the operation of the retail deposit market, which we presently do not stress. For a given difference in the local deposit base, as captured by $M_A - M_B = 2z > 0$, the respective difference in attracted retail deposits, $R_A - R_B < M_A - M_B$, is strictly higher when banks' lending markets are less correlated (lower ρ). Interestingly, however, as the difference in the deposit base z increases, while interbank lending is always strictly increasing, this may not hold for the volume of retail deposits that bank B attracts in market A .

under retail deposit competition.

Proof. First, for

$$\bar{s} \geq \frac{M+z}{M-z} [L'(M-z) - 1],$$

there is no reallocation on the retail deposit market but interbank lending when $[p + \rho(1-p)] > \frac{L'(M+z)}{L'(M-z)}$. Next, for

$$L'(M+z) - 1 \leq \bar{s} < \frac{M+z}{M-z} [L'(M-z) - 1],$$

there will be interbank lending when retail deposit competition without subsequent interbank lending leads to an allocation where $[p + \rho(1-p)] > \frac{L'(R_A)}{L'(R_B)}$. Finally, for

$$\bar{s} < L'(M+z) - 1,$$

interbank lending is never optimal from the bank's perspective and the allocation is not changed by introducing the potential for interbank lending. **Q.E.D.**

2 Nash Bargaining over Interbank Lending

We will now relax the assumption that the creditor bank A can make a take-it-or-leave-it offer to the debtor bank B and suppose instead that the surplus generated by interbank lending is shared according to axiomatic Nash bargaining.

For this recall first that there is no asymmetric information, so that bargaining proceeds under common knowledge. When banks do not come to an agreement, we denote their respective outside options by π_{n0} , which are obtained by substituting $W = 0$ and $w = 0$ into (3). We next derive the bargaining (or Pareto) frontier. For some given (feasible) value of profits $\tilde{\pi}_{n'}$ for bank n' this entails finding the interbank loan that maximizes the other bank's profits, π_n , subject to the constraint that $\pi_{n'} \geq \tilde{\pi}_{n'}$. It is inessential which banks we choose as n or n' . For specificity, suppose we maximize π_A . As is immediate, the constraint $\pi_B \geq \tilde{\pi}_B$ will be binding so that the maximization problem gives rise to a function $\pi_A = \psi(\pi_B = \tilde{\pi}_B)$. There are two cases to distinguish. In the first case, interbank lending is not optimal from the banks' shareholders' perspective, so that there does not exist a pair (π_A, π_B) with $\pi_A \geq \pi_{A0}$ and $\pi_B \geq \pi_{B0}$, where at least one holds strictly. In the second

case, such a pair exists. Then, if ψ is concave (it is, in fact, linear, as we show below), the symmetric Nash solution is characterized as follows: The uniquely obtained solution (π_A, π_B) maximizes the (symmetric) Nash product $[\psi(\pi_B) - \pi_{A0}][\pi_B - \pi_{B0}]$, which from the first-order condition is the case if

$$\frac{\psi(\pi_B) - \pi_{A0}}{\pi_B - \pi_{B0}} = -\psi'(\pi_B). \quad (\text{A9})$$

The derivation of the Nash bargaining solution - or, more precisely, the derivation of the frontier $\pi_A = \psi(\pi_B)$ - is complicated by the following feature. The bargaining frontier does not have slope of minus one, as in the most standard case where risk-neutral players can simply make a fixed transfer ("transferable utility"). This results from the fact that a debtor bank can make its contractual repayment only if its own corporate loans perform. And even when a creditor bank receives such payment, it may go straight to depositors rather than banks' shareholders when the bank becomes insolvent. It is now convenient to solve explicitly for w from the binding constraint $\pi_B \geq \tilde{\pi}_B$, so that in this case

$$w = L(R_B + W) - R_B(1 + r_B) - \tilde{\pi}_B \frac{1}{p}. \quad (\text{A10})$$

Hence, in case there is a loan of size W from bank A to bank B , then the repayment w as specified in (A10) ensures that bank B 's profits are just equal to $\tilde{\pi}_B$. Substituting for w , $W \geq 0$ is then chosen so as to maximize

$$\begin{aligned} \pi_A = & [p^2 + \rho p(1 - p)] \left[L(R_A - W) - R_A(1 + r_A) + L(R_B + W) - R_B(1 + r_B) - \tilde{\pi}_B \frac{1}{p} \right] \\ & + p(1 - p)(1 - \rho) \left[\begin{aligned} & \max \{0, L(R_A - W) - R_A(1 + r_A)\} \\ & + \max \left\{ 0, L(R_B + W) - R_B(1 + r_B) - \tilde{\pi}_B \frac{1}{p} - R_A(1 + r_A) \right\} \end{aligned} \right]. \end{aligned}$$

Importantly, this implies that $\tilde{\pi}_B$ does not affect the optimal choice of W .

As a consequence, we have that under Nash bargaining the same optimal W obtains as when one bank can make a take-it-or-leave-it offer. That the corresponding repayment level w , as given by (A10), is different does not affect our results qualitatively (albeit it affects the thresholds for z and ρ got which Cases 1 and 2 apply in equilibrium).